

Rotation Matrices Part 1

EGR 445/545

GVSU

Notes

- clean up Rz derivation stuff
 - move some to review
 - get rid of some
- add in stuff about multiple rotations
- and HT preview and side view content to LA 1 review
- you also need some content on finding rotation matrices by inspection
 - I guess there is some

Nothing to see
here

Plan for ~~Tonight~~ today

- review:
 - LA 01
 - R_z derivation
 - columns are unit vectors
- R_y derivation (you)
- multiple rotations
- rotation matrices by inspection
 - multiple/compound rotations
- the transpose is the inverse of a rotation matrix
- learning activities 2 and 3

Thanks

- thanks for a good first class
- I think we are off to a solid start
 - everyone has jupyter running and my robotics module installed (I think)
- I commend you
 - this is not an easy course, but you chose it

Quick Review

- what did we learn last time?
 - what is the primary challenge in industrial robotics?
 - why do we need to learn robot kinematics?
 - what is a rotation matrix and where do they come from?

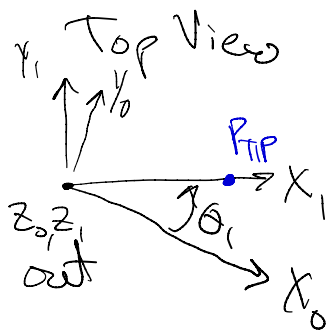
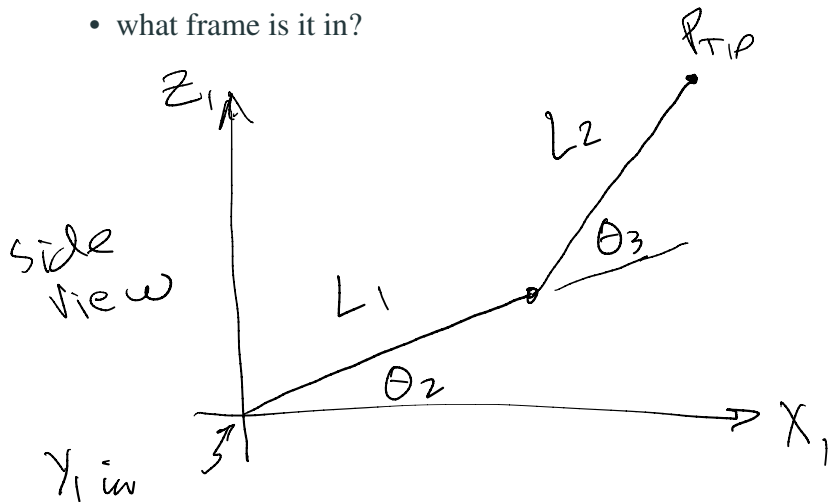
} find tip position
and orientation
from joint
angles

→ Models orientation of coord. sys

→ Signs matter and are tricky
for rotation matrices

LA 1 Review: Side view

- what can we learn from a side view?
- what frame is it in?



Side view

y_1 in

z_1

x_1

${}^2P_{TIP}$

P_{TIP}

θ_3

θ_2

L_0

${}^2P_{TIP}$

$${}^3P_{TIP} = \begin{bmatrix} L_2 \\ 0 \\ 0 \end{bmatrix}$$

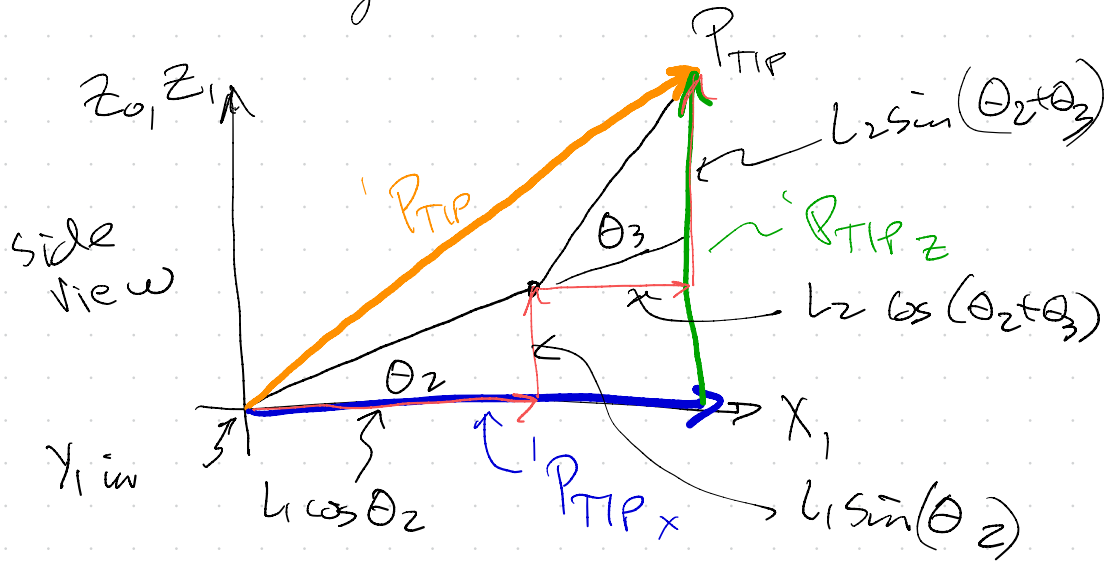
$${}^2R = R_y(-\theta_3)$$

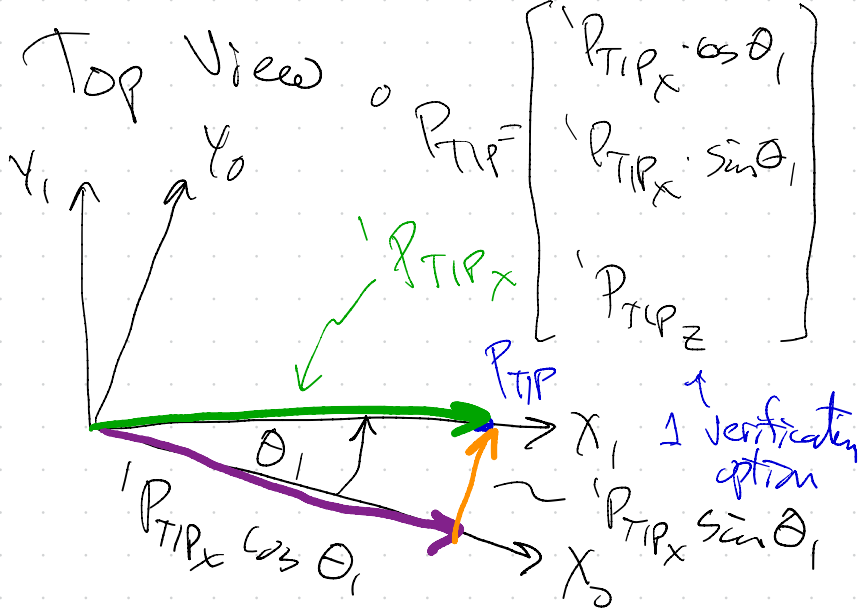
$$\begin{bmatrix} L_1 \\ 0 \\ 0 \end{bmatrix}$$

$${}^2R \cdot {}^3P_{TIP}$$

$${}^1R = R_y(-\theta_2)$$

Trig Verification





LA 1 Review: Top view

- how does a top view help us get to a trig answer check for ${}^0P_{tip}$?

see previous slide

LA 1: Intro to HT Matrices

for ${}^2P_{tip}$, most people did this:

$${}^2P_{tip} = {}^2_3\mathbf{R} \cdot {}^3P_{tip} + \begin{bmatrix} L_1 \\ 0 \\ 0 \end{bmatrix}$$

but can we do this in one operation?

$$\begin{bmatrix} {}^2P_{tip} \\ 1 \end{bmatrix} = \begin{bmatrix} {}^2_3\mathbf{R} & L_1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} {}^3P_{tip} \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} {}^2P_{TLP} \\ 1 \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} & L_1 \\ r_{21} & r_{22} & r_{23} & 0 \\ r_{31} & r_{32} & r_{33} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} L_2 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

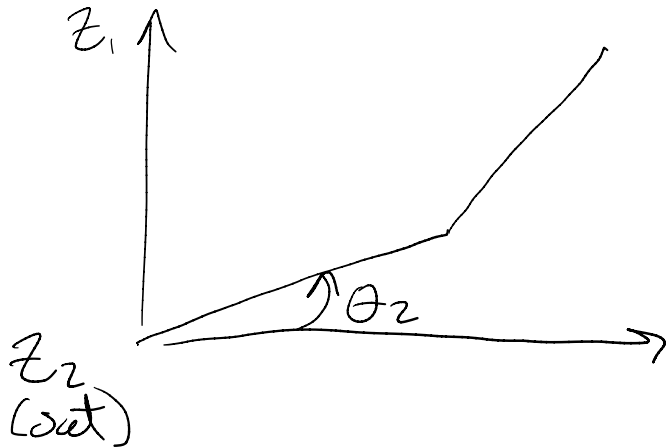
$$\begin{bmatrix} {}^2P_{TLP} \\ 1 \end{bmatrix} = \begin{bmatrix} L_2 r_{11} + L_1 \\ L_2 r_{21} \\ L_2 r_{31} \\ 1 \end{bmatrix} = {}^2R^{-3}P_{TLP} + \begin{bmatrix} L_2 \\ 0 \\ 0 \end{bmatrix}$$

$${}^0 P_{TIP} = {}^0 T_1 \cdot {}^1 T_2 \cdot {}^2 T_3 \cdot {}^3 P_{TIP}$$

LA 1: DH Preview

How could we make all of the θ_i rotations about Z axes?

- **hint:** Z_1 and Z_2 are not parallel, so we will need an intermediate frame



$${}^1_2R = R_x(90) \cdot R_z(\theta_2)$$

$${}^{1.5}_1R = R_x(90)$$

$${}^{1.5}_2R = R_z(\theta_2)$$

$${}^1_2R = {}^{1.5}_1R \cancel{{}^{1.5}_2R}$$

Quick Segue: Multiple Rotations

- let the notation guide you
- more later

See previous
slide

Review of Rotation Matrices

Understanding Rotation Matrices

What does this equation mean?:

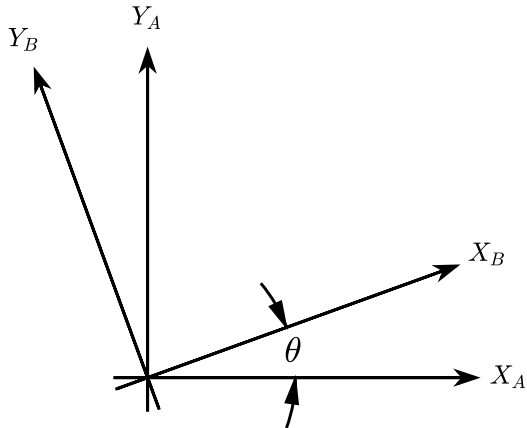
$${}^A_B\mathbf{R} = \begin{bmatrix} {}^A\hat{X}_B & {}^A\hat{Y}_B & {}^A\hat{Z}_B \end{bmatrix} = \begin{bmatrix} \hat{X}_B \cdot \hat{X}_A & \hat{Y}_B \cdot \hat{X}_A & \hat{Z}_B \cdot \hat{X}_A \\ \hat{X}_B \cdot \hat{Y}_A & \hat{Y}_B \cdot \hat{Y}_A & \hat{Z}_B \cdot \hat{Y}_A \\ \hat{X}_B \cdot \hat{Z}_A & \hat{Y}_B \cdot \hat{Z}_A & \hat{Z}_B \cdot \hat{Z}_A \end{bmatrix}$$

component of $\hat{X}_B$
along $\hat{X}_A$

columns of ${}^A_B\mathbf{R}$ are the
frame B unit vectors expressed
in frame A

Rotation Matrix Sign Convention

- we rotate from A to B , but map from B to A
 - start with frames A and B aligned/coincident
 - rotate frame B away from A in a right-hand sense
 - ${}^A_B\mathbf{R}$ maps a vector from frame B back to frame A



Signs are important

- you have a growing suspicion that memorizing or googling rotation matrix is very risky as far as signs go
 - the same applies when using them correctly
- getting the signs right can be tricky when it comes to rotation matrices
- if you understand where rotation matrices come from, you will be protected from these sign errors

Signs are important - continued

Ginsberg:

$$[R_x] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta_x & \sin \theta_x \\ 0 & -\sin \theta_x & \cos \theta_x \end{bmatrix}$$

$$[R_y] = \begin{bmatrix} \cos \theta_y & 0 & -\sin \theta_y \\ 0 & 1 & 0 \\ \sin \theta_y & 0 & \cos \theta_y \end{bmatrix}$$

$$[R_z] = \begin{bmatrix} \cos \theta_z & \sin \theta_z & 0 \\ -\sin \theta_z & \cos \theta_z & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Craig:

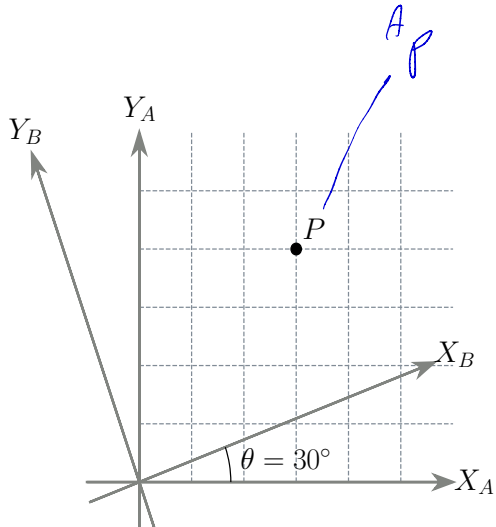
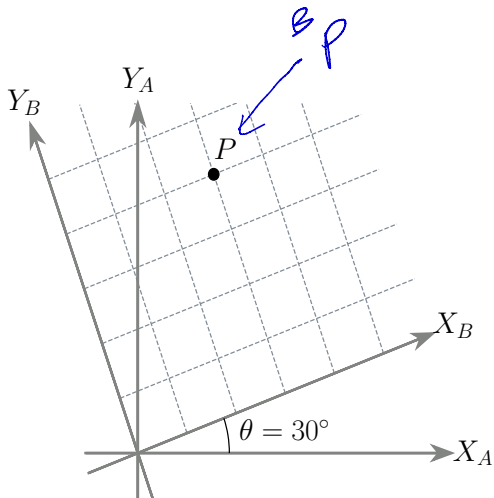
$$R_X(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix}$$

$$R_Y(\theta) = \begin{bmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{bmatrix}$$

$$R_Z(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

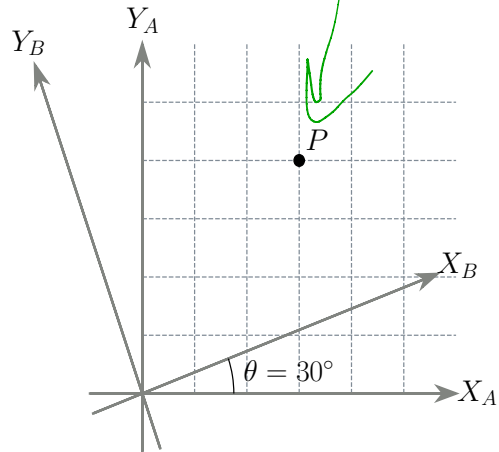
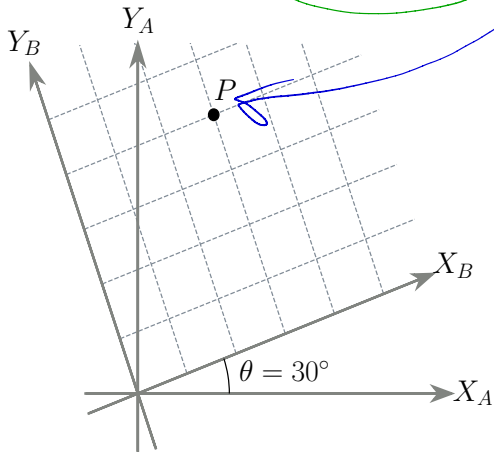
These are two different problems

- what's the difference?

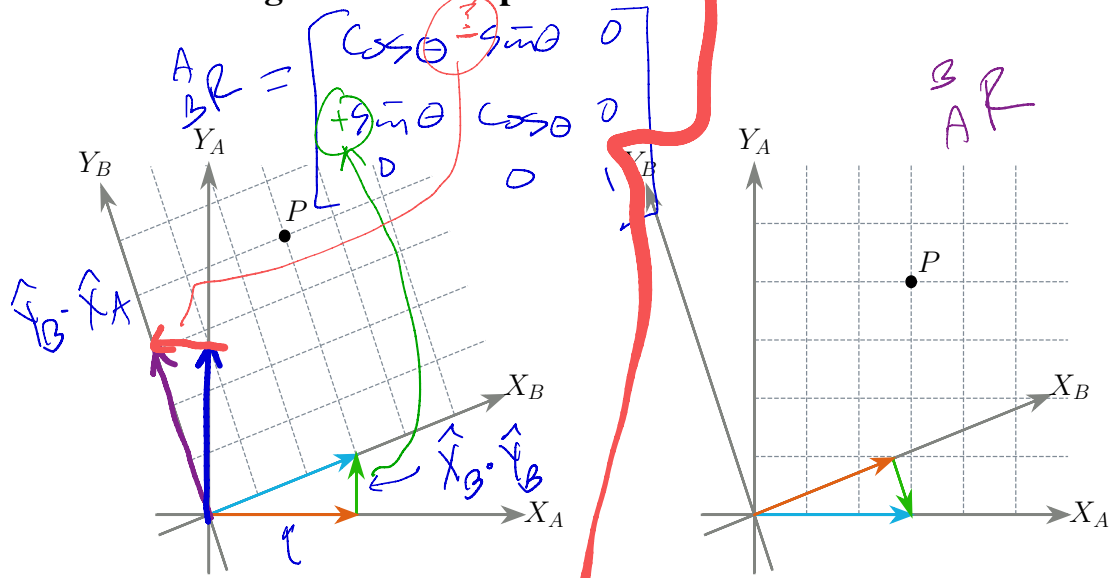


Which is which?

$${}^B P = {}^B \mathbf{R} \cdot {}^A P \text{ vs. } {}^A P = {}^A \mathbf{R} \cdot {}^B P$$



What are the signs of the components in the other frame?



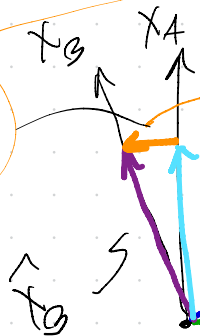
Review Exercise: Derive R_y

- draw a sketch and come up with an expression for ${}^A_B\mathbf{R}$ for a positive rotation about the Y axis

$${}^A\mathcal{P} = {}^A_B\mathcal{R} \cdot {}^B\mathcal{P}$$

R_y derivation

$$\hat{x}_B \cdot \hat{z}_A$$



$${}^A_B R = R_y =$$

$$\begin{bmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{bmatrix}$$

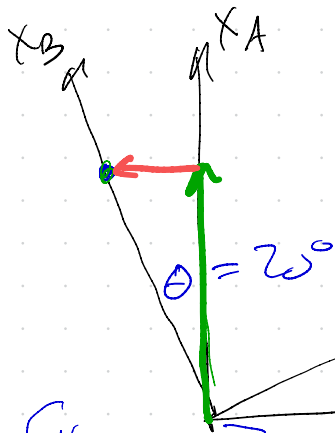
$$\hat{z}_B \cdot \hat{x}_A$$

$$\hat{z}_B \cdot \hat{x}_A$$

$\downarrow$ A and B out

eqn. 2.78

$${}^A P = \begin{bmatrix} 4 \cos 20^\circ \\ 0 \\ -4 \sin 20^\circ \end{bmatrix}$$



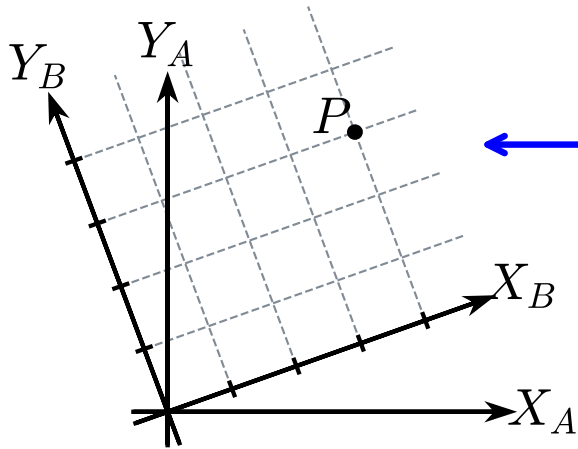
$${}^B P = \begin{bmatrix} 4 \\ 0 \\ 0 \end{bmatrix}$$

Conventions for rotations and mappings

- start with frames A and B coincident
- frame B rotates by angle θ
 - we rotate from A to B
 - follow the right hand rule
- ${}^A_B\mathbf{R}$ maps from frame B back to frame A
- this might seem weird at first
- it corresponds to what we know and what we want to find
 - when $\theta = 0$, frames A and B are aligned
 - when we actuate the joint, frame B moves
 - we ultimately know P_{tip} in the last frame and want to map back to frame 0

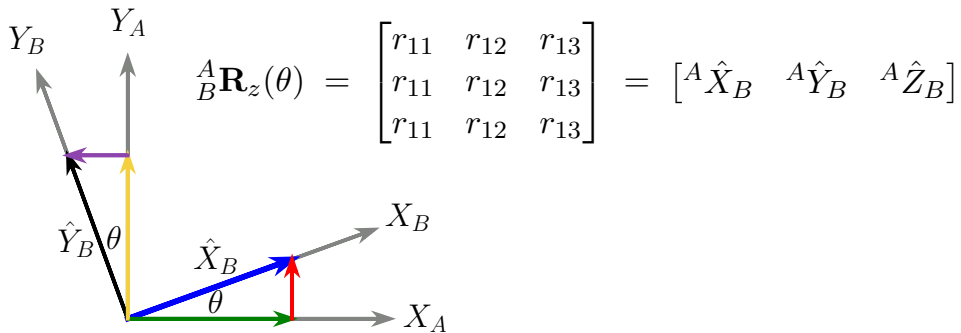
R_z Derivation Review

Rotation Matrix Intro

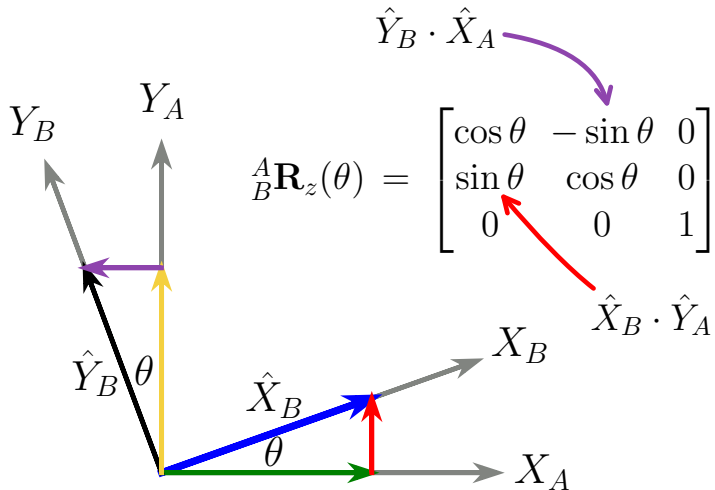


$${}^A P = {}^A_B R {}^B P$$

Rz Derivation Overview



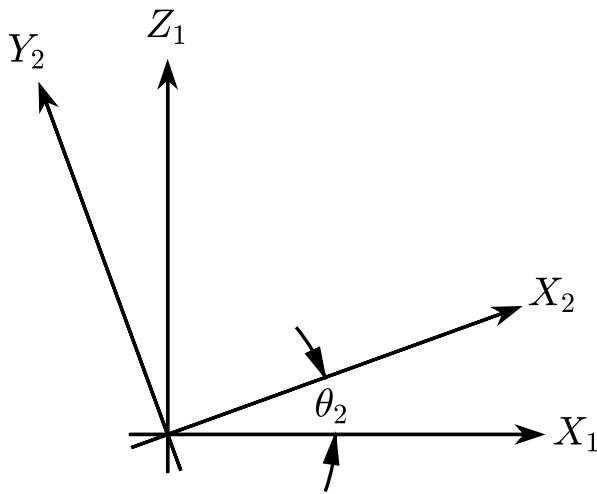
R_z Derivation Summary



Multiple Rotations

Multiple Rotations

- why can't we easily find ${}^1_2\mathbf{R}$ with one rotation?



Multiple Rotations

- if it takes two rotations to get from frame A to frame B , define an intermediate frame I
- first rotate from frame A to frame I
- then rotate from frame I to frame B
- the combined rotation matrix will be

$${}^A_B\mathbf{R} = {}^A_I\mathbf{R} \cdot {}^I_B\mathbf{R}$$

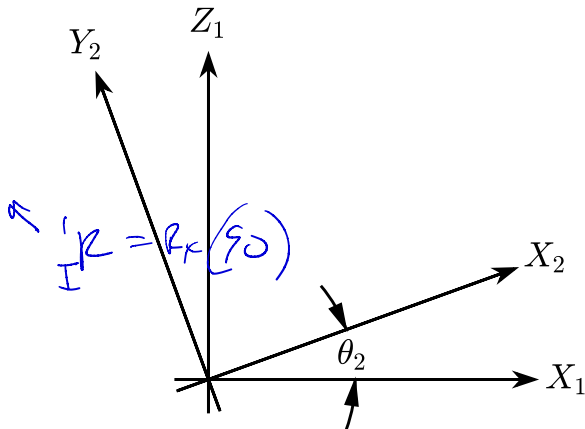
$\sim R_1(\theta_1)$

$\leftarrow R_2(\theta_2)$

Multiple Rotations: ${}^1_2\mathbf{R}$ for the Plexiglass Robot

Do the θ_2 rotation second

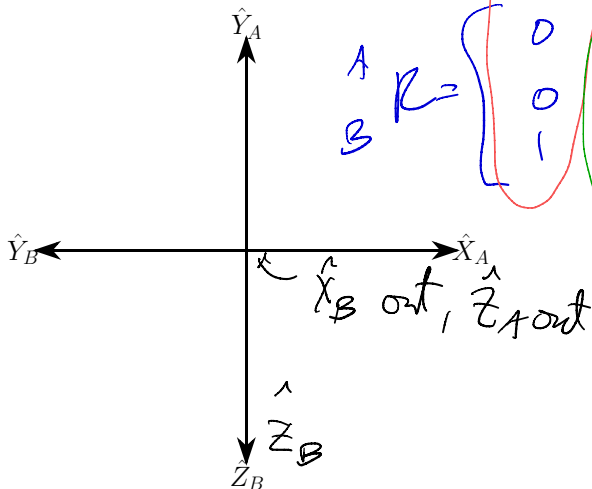
- we want θ_2 to rotate about $\hat{Z}_2$
- what would the first rotation be?
 - how do we get $\hat{Z}_2$ pointing where we need?
- where would the intermediate axes point?



Rotation Matrices by Inspection

Finding ${}^A_B\mathbf{R}$ with Arbitrary Orientation

What is ${}^A_B\mathbf{R}$ for this sketch?



$${}^A_B\mathbf{R} = \begin{bmatrix} \hat{X}_B & \hat{Y}_B & \hat{Z}_B \end{bmatrix}$$

$${}^A_B\mathbf{R} = \begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & -1 \\ 1 & 0 & 0 \end{bmatrix}$$

The Inverse of a Rotation Matrix

Dot Products and Rotation Matrices

What does this mean? (eqn 2.3):

$${}^A_B\mathbf{R} = \begin{bmatrix} {}^A\hat{X}_B & {}^A\hat{Y}_B & {}^A\hat{Z}_B \end{bmatrix} = \begin{bmatrix} \hat{X}_B \cdot \hat{X}_A & \hat{Y}_B \cdot \hat{X}_A & \hat{Z}_B \cdot \hat{X}_A \\ \hat{X}_B \cdot \hat{Y}_A & \hat{Y}_B \cdot \hat{Y}_A & \hat{Z}_B \cdot \hat{Y}_A \\ \hat{X}_B \cdot \hat{Z}_A & \hat{Y}_B \cdot \hat{Z}_A & \hat{Z}_B \cdot \hat{Z}_A \end{bmatrix}$$

The Transpose is the Inverse

- what do we know about the columns of ${}^A_B\mathbf{R}$?
 - what does this tell us about ${}^A_B\mathbf{R}^T$?

Dot Products Part 2

$${}^A_B\mathbf{R} = \begin{bmatrix} {}^A\hat{X}_B & {}^A\hat{Y}_B & {}^A\hat{Z}_B \end{bmatrix} \text{ (unit vectors as columns)}$$

$${}^A_B\mathbf{R}^T = \begin{bmatrix} {}^A\hat{X}_B^T \\ {}^A\hat{Y}_B^T \\ {}^A\hat{Z}_B^T \end{bmatrix} \text{ (unit vectors as rows)}$$

Handwritten annotations showing the dot products of the unit vectors:

$$\begin{matrix} {}^A\hat{X}_B^T \cdot {}^A\hat{X}_B & {}^A\hat{X}_B^T \cdot {}^A\hat{Y}_B & {}^A\hat{X}_B^T \cdot {}^A\hat{Z}_B \\ {}^A\hat{Y}_B^T \cdot {}^A\hat{X}_B & {}^A\hat{Y}_B^T \cdot {}^A\hat{Y}_B & {}^A\hat{Y}_B^T \cdot {}^A\hat{Z}_B \\ {}^A\hat{Z}_B^T \cdot {}^A\hat{X}_B & {}^A\hat{Z}_B^T \cdot {}^A\hat{Y}_B & {}^A\hat{Z}_B^T \cdot {}^A\hat{Z}_B \end{matrix}$$

$${}^A_B\mathbf{R}^T \cdot {}^A_B\mathbf{R} = ?$$

Transpose as Inverse Conclusion

$${}^A_B\mathbf{R}^T = \begin{bmatrix} {}^A\hat{X}_B^T \\ {}^A\hat{Y}_B^T \\ {}^A\hat{Z}_B^T \end{bmatrix} \quad \text{and} \quad {}^A_B\mathbf{R} = \begin{bmatrix} {}^A\hat{X}_B & {}^A\hat{Y}_B & {}^A\hat{Z}_B \end{bmatrix}$$

$${}^A_B\mathbf{R}^T \cdot {}^A_B\mathbf{R} = \begin{bmatrix} {}^A\hat{X}_B^T \cdot {}^A\hat{X}_B & {}^A\hat{X}_B^T \cdot {}^A\hat{Y}_B & {}^A\hat{X}_B^T \cdot {}^A\hat{Z}_B \\ {}^A\hat{Y}_B^T \cdot {}^A\hat{X}_B & {}^A\hat{Y}_B^T \cdot {}^A\hat{Y}_B & {}^A\hat{Y}_B^T \cdot {}^A\hat{Z}_B \\ {}^A\hat{Z}_B^T \cdot {}^A\hat{X}_B & {}^A\hat{Z}_B^T \cdot {}^A\hat{Y}_B & {}^A\hat{Z}_B^T \cdot {}^A\hat{Z}_B \end{bmatrix}$$

$${}^A_B\mathbf{R}^T \cdot {}^A_B\mathbf{R} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$${}^A_B\mathbf{R}^T = {}^A_B\mathbf{R}^{-1}$$

Learning Activity 2

- practice with rotation matrices

Learning Activity 3

Numeric verification of

$${}^B_A\mathbf{R} = {}^A_B\mathbf{R}^T = {}^A_B\mathbf{R}^{-1}$$